NEW CURRICULUM ASSESSMENT GUIDELINES FOR PHYSICS AT ADVANCED SECONDARY LEVEL

ASSESSMENT OBJECTIVES

The end of Cycle assessment for Physics will be guided by four assessment objectives (AO) focusing on the learner's ability to:

AO1: Evaluate how forces affect static, moving bodies and their structures and use this knowledge to design safe, efficient, and practical solutions in everyday situations

AO2: Investigate how energy is transferred and transformed, design devices and systems used in daily life, industry, and technology that employ energy transformations.

AO3: Explore how electric and magnetic fields interact in systems and devices, and use this understanding to explain and model technologies for power transmission, energy storage, and electronic control in daily life and industry.

AO4: Investigate atomic, quantum, and nuclear phenomena, and use this understanding to explain how modern technologies work and to assess the benefits and risks of their applications.

LINKAGE BETWEEN ASSESSMENT OBJECTIVES, CONSTRUCTS AND SYLLABUS

Assessment Objective	Construct	Construct Description	Topics in the syllabus
AO1	Force and Motion	Analysis of how force affects the state of bodies	1. Measurement and Dimensions of Physical Quantities 2. Statics 3. Linear Motion 4. Motion Under Gravity 5. Work, Energy and Power

			6. Solid Friction 7. Fluid Mechanics 8. Mechanical Properties of Matter 20. Circular Motion 22. Gravitation
AO2	Energy	Exploration and application of energy forms and transformations.	9. Thermometry 10. Heat Quantities 11. Transfer of Heat 12. Behavior of Gases 13. Thermodynamics 14. Reflection of Light 15. Refraction of Light 16. Optical Instruments 21. Simple Harmonic Motion 23. Progressive Waves 24. Stationary Waves 25. Sound Waves
AO3	Charges and Fields	Exploration of how electric and magnetic fields interact in systems and devices.	17. Electrostatics 18. Capacitors 19. Digital Electronics 26. Current Electricity 27. Magnetism in Matter 28. Magnetic Effect of an electric Current 29. Electromagnetic Induction 30. A.C Circuits
AO4	Particles	Investigation of atomic, quantum, and nuclear phenomena and their applications	31. Atomic Particles 32. Quantum Theory 33. Nuclear Processes

STRUCTURE OF THE EXAMINATION PAPERS

There will be two examination papers for Physics at UACE

PAPER 1 (THEORY PAPER)

This will contain four sections A, B, C and D. The items will be scenario based.

Section A

This will comprise of one item from construct four; Particles addressing assessment objective 4.

Section B

This will comprise of two items from construct~~s~~ one; Force and Motion addressing assessment objective 1

Section C

This will comprise of two items from construct two; Energy addressing assessment objective 2

Section D

This will comprise of two items from construct three; Charges and Fields addressing assessment objective 3

NUMBER OF QUESTIONS ATTEMPTED

The learner will attempt four items, including the one compulsory item from section A and one item from each of the sections B, C and D

TIME

The entire paper will take 2hours and 40 minutes.

PAPER 2 (PRACTICAL PAPER)

This will contain **two items**.

Items in this paper will come from any of the **four** constructs.

The items in the paper will be scenario based

The paper will take **2hours and 45** minutes

The learner will attempt only **one item**.

END